Olympiad Practice Examination

Eleven Topics — One Problem Each

Suggested time: 4 hours. Full solutions are expected. Problems are not necessarily in order of difficulty. Analytic and synthetic geometry are treated as separate topics. Calculators are not needed.

Problem 1 [Polynomials]

Determine all nonconstant polynomials $P(x)$ with real coefficients such that

$$(x-1)P(x+1) = (x+2)P(x) \qquad \text{for every } x \in \mathbb{R}.$$

Problem 2 [Inequalities]

Let $a, b, c > 0$ satisfy $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

Determine all equality cases.

Problem 3 [Functional Equations]

Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(x+y) + f(x-y) = 2f(x)\cos y \qquad \text{for all } x, y \in \mathbb{R}.$$

Problem 4 [Complex Numbers]

Let $n \geq 2$ be an integer, and let $\omega = e^{2\pi i/n}$. Evaluate, in closed form,

$$\sum_{k=0}^{n-1} \frac{1}{2-\omega^k}.$$

Problem 5 [Combinatorics]

For an integer $n \geq 1$, a sequence of n symbols $+$ and n symbols $-$ is called *balanced* if every nonempty initial segment contains at least as many $+$ symbols as $-$ symbols. Prove that the number of balanced sequences is the Catalan number

$$\frac{1}{n+1}\binom{2n}{n}.$$

Problem 6 [Pigeonhole Principle]

Let $n \geq 1$ be an integer. Determine the greatest possible number of elements in a subset S of $\{1, 2, \ldots, 2n\}$ such that no element of S divides a different element of S.

Problem 7 [Graph Theory]

A connected graph has $n \geq 2$ vertices, and every vertex has even degree. Prove that its edges can be oriented so that, at every vertex, the indegree and outdegree are equal.

Problem 8 [Game Theory]

Two players alternately remove stones from a pile initially containing 2026 stones. On a turn, a player must remove 2^j stones for some nonnegative integer j, provided that many stones remain. The player who removes the last stone wins. Determine which player has a winning strategy and describe that strategy completely.

Problem 9 [Number Theory]

Determine all positive integers n for which

$$2^n + n \mid 8^n + n.$$

Problem 10 [Geometry — Analytic]

The parabola $y = x^2$ has focus $F = (0, \frac{1}{4})$. A line through F meets the parabola at two distinct points A and B. Prove that the circle with diameter AB is tangent to the line $y = -\frac{1}{4}$.

Problem 11 [Geometry — Synthetic]

Let $ABCD$ be a convex cyclic quadrilateral whose diagonals AC and BD intersect at P and are perpendicular. The line through P perpendicular to AB meets line CD at M. Prove that M is the midpoint of CD.